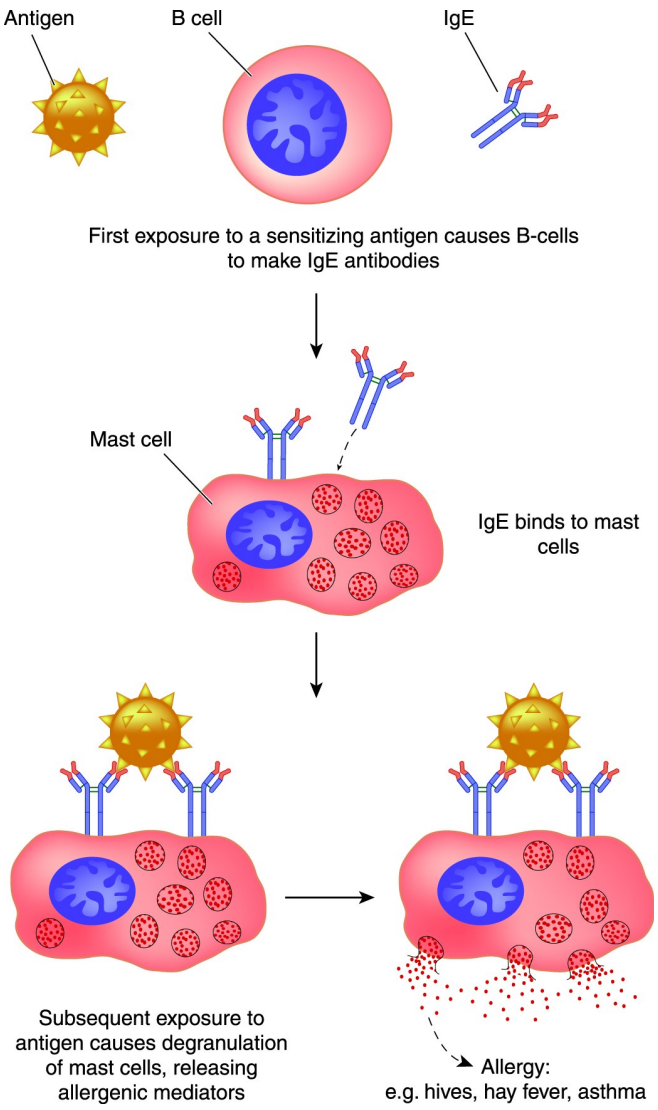
Anaphylaxis is a life-threatening reaction that occurs when a patient is previously exposed to an antigen, develops antibodies to it, then reacts to it on subsequent exposure with massive degranulation of mast cells and basophils in an IgE-mediated chain reaction. This is in contrast to cell-mediated immunity in which the antigen is presented to the T lymphocyte by infected cells or by 'antigen-presenting cells'.

The initial reaction occurs within 5 to 10 minutes of exposure. The mast cells and basophils release histamine and other vasoactive mediators that are primarily responsible for the urticaria, pruritus, and vascular collapse. The hypotension seen in these patients is due to increased capillary permeability which allows as much as 50% of the intravascular volume to extravasate into the extracellular space. Histamine also results in increased heart rate and myocardial contractility Laryngeal edema, bronchospasm, and hypoxemia may accompany the reaction.

Diagnosis of anaphylaxis may be difficult in the operating room as anesthesia drugs can alter vasodilating mediators and delay the recognition of the reaction. It may also resemble pulmonary embolus, myocardial infarction, or acute aspiration.

The first-line treatment of anaphylaxis is several liters of crystalloid and epinephrine in doses beginning at 10 to 100 mcg IV. By increasing intracellular cyclic adenosine monophosphate (cAMP), epinephrine restores normal capillary permeability and helps relax bronchial smooth muscle. Antihistamines reduce some symptoms such as pruritus and bronchospasm, but is unable to reverse the negative inotropic symptoms mediated by leukotrienes. Beta-agonists are helpful in treating bronchospasm and corticosteroids inhibit the release of leukotriene-mediated arachidonic acid and may be especially helpful in allergic reaction caused by activation of the complement system.



Most intraoperative drug reactions (60%) are due to muscle relaxants. About 15% are due to latex. Antibiotics are responsible for 5-10%. Opioids are responsible for less than 5% of intraoperative drug reactions.

Anaphylactoid reactions differ from anaphylactic reactions in that they do not involve IgE antibodies, but are instead mediated by mast cell and basophil activation.

Hines RL, Marschall KE. Stoeltings Anesthesia & Co-Existing Diseases. 6th ed. New York, NY: Churchill-Livingston; 2012: 523-525.

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